PF – Group D Semester Project Proposal

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Pseudo E-Challan System — Concept & Roadmap

1. Overview

This document provides a high-level conceptual explanation of a pseudo e-challan system designed entirely in C++ using OpenCV and file-based storage. The system simulates the behavior of traffic violation detection, challan generation, user notification, and payment updates. It is designed as a proof-of-concept and does not require SQL or external sensors other than a camera.

2. System Components

The system consists of several interconnected functional modules, each handling a specific aspect of the workflow:

• Input Manager — Handles live camera feed or static image input.

• Image Processing Unit — Detects vehicles and extracts number plates using OpenCV.

• Violation Decision Engine — Applies rule-based checks for identifying violations.

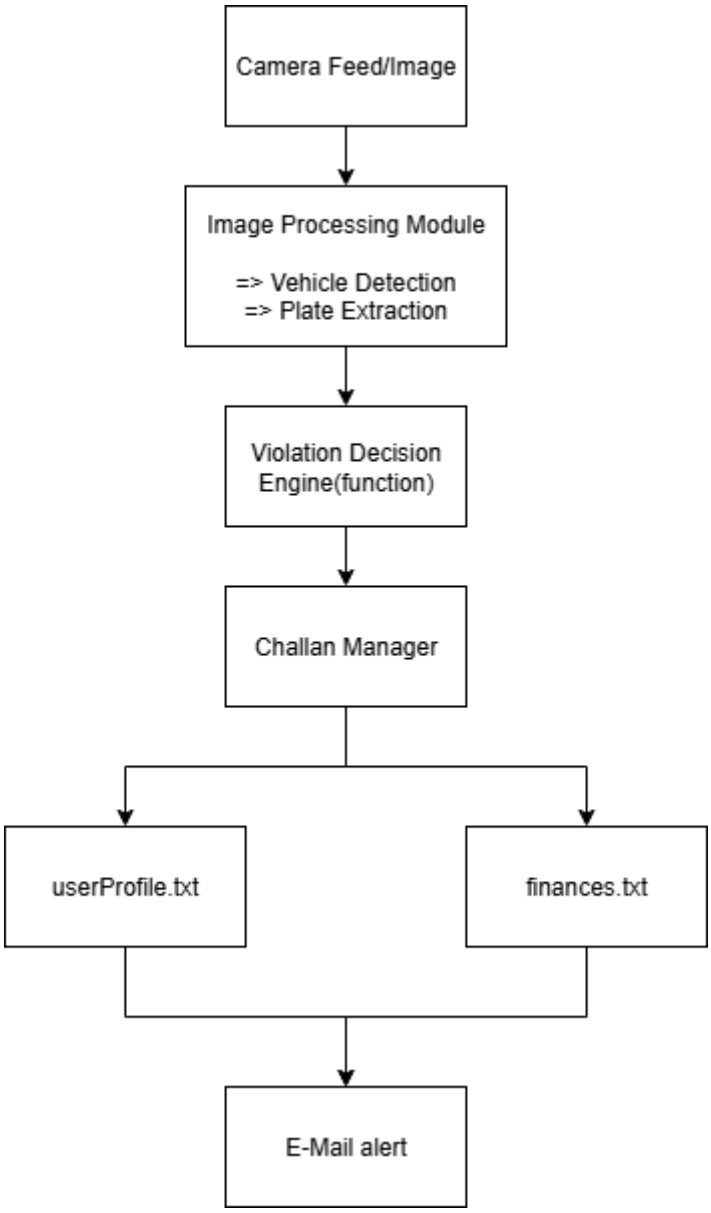
• Challan Manager — Creates and updates challan records stored in text files.

• User Database Manager — Maintains vehicle and owner information in file form.

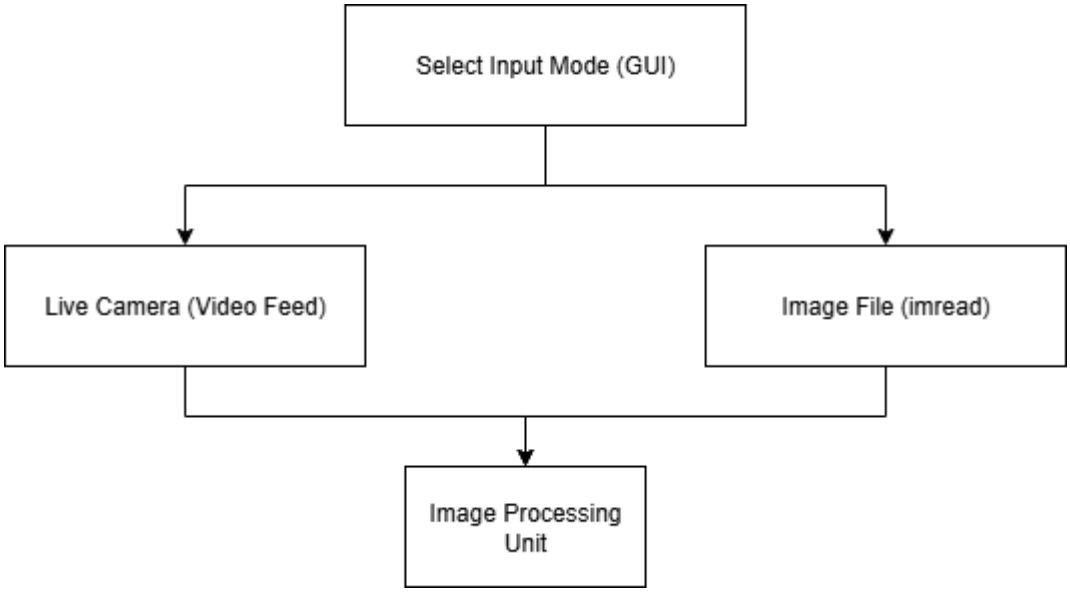
• Finance Manager — Simulates user account balance and challan payments.

• Email Notification Module — Sends challan alerts through SMTP.

3. High-Level Architecture Diagram



4. Input Mode Switching Diagram



5. Roadmap for Development

Stage 1 — Core Architecture Setup

• Set up project structure in C++.

• Implement file-based data storage.

• Build basic user interface for mode selection.

Stage 2 — Image Input & Detection

• Implement support for live camera input.

• Implement static image testing.

• Develop number plate detection using OpenCV contour techniques.

Stage 3 — Violation Detection Logic

• Define rule-based violation categories.

• Integrate bounding-box detection and plate localization.

• Trigger challan creation based on detected violations.

Stage 4 — Challan & Profile Management

• Create text-file-based challan logs.

• Create and manage user vehicle profile database.

• Add challan update and retrieval options.

Stage 5 — Finance & Payment Simulation

• Implement payment system using file-stored balances.

• Update challan status from UNPAID to PAID.

• Track transaction history in files.

Stage 6 — Email Communication System

• Implement SMTP-based email alert system.

• Notify vehicle owners of challan generation.

• Add configurable message templates.

Stage 7 — Testing & Validation

• Test system using multiple image samples.

• Test live camera scenarios.

• Validate challan creation and payment workflows.

6. Conclusion

This system allows a complete yet simplified demonstration of traffic violation detection and e-challan workflow in C++. It makes use of only camera input, OpenCV processing, and file-based databases, providing a clean and flexible development path for a POC application.